



The p-Block Elements

* "The p-block elements exhibit a wide range of physical and chemical properties due to variable oxidation states, bonding ability and catenation."

1. GENERAL INTRODUCTION

- The p-block elements are those in which the last electron enters into p-orbital.
- These elements are in groups 13 to 18 of the periodic table except He.
- General electronic configuration:
 $ns^2 np^{1-6}$
- They show a variety of oxidation states ranging from -4 to +7.
- These elements form covalent compounds predominantly.

P-BLOCK IN PERIODIC TABLE

1	2	13	14	15	16	17	18
1							He

Groups →

Periods →

p-block

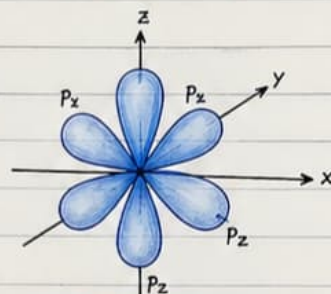
Groups 13 to 18
Periods 2 to 6

2. ELECTRONIC CONFIGURATIONS & CHARACTERISTICS

- Group 13: $ns^2 np^1$ (e.g., B, Al)
- Group 14: $ns^2 np^2$ (e.g., C, Si)
- Group 15: $ns^2 np^3$ (e.g., N, P)
- Group 16: $ns^2 np^4$ (e.g., O, S)
- Group 17: $ns^2 np^5$ (e.g., F, Cl)
- Group 18: $ns^2 np^6$ (except He)
(e.g., Ne, Ar)

Salient Features

- The size increases down the group.
- Ionization enthalpy generally decreases down the group.
- Electronegativity decreases down the group, but increases across the period.
- $p\pi-p\pi$ multiple bonding is possible.
- Catenation is prominent.



Three p-orbitals
along x, y, z axes

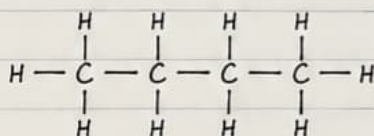
3. IMPORTANT CONCEPTS

(i) Catenation

- Ability of an element to form bonds with itself to form long chains or rings.

Example:

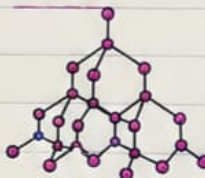
- Catenation is maximum in carbon.



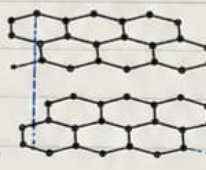
(ii) Allotropy

- Existence of an element in two or more forms in the same physical state.

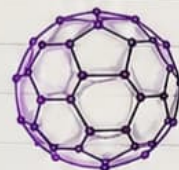
Example: Carbon → Diamond, Graphite, Fullerene



Diamond



Graphite



Fullerene

(iii) Variable Oxidation States

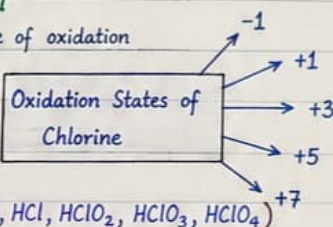
- p-block elements show a wide range of oxidation states.

Example:

+1: Tl, +2: Pb, Sn

+4: Sn, Pb

+7: Cl, Mn



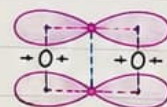
(iv) $p\pi-p\pi$ Multiple Bonding

- Stronger multiple bonds between p-orbitals is possible.

Example:

O_2 molecule (double bond)

$\text{O}=\text{O}$ ($p\pi-p\pi$ bond)



p-Block Elements → Versatile nature due to electronic configuration, catenation, multiple bonding & variable oxidation states.



Chapter-10: The p-Block Elements

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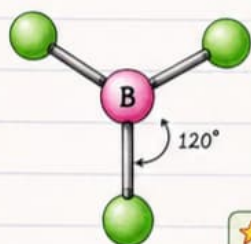
4. GROUP 13 ELEMENTS

- General configuration: $ns^2 np^1$
- Common oxidation state: +3 (most stable)
- The +1 state becomes more stable down the group due to inert pair effect.

Examples:

(i) Boron (B)

- Electron deficient and forms compounds with covalent character.
- Forms electron-deficient compounds like BCl_3 .



BCl_3 (Trigonal planar)
 sp^2 hybridisation
Electron deficient
(6 electrons around B)

★ Boron does not form oxides with water.

Trends down the group:

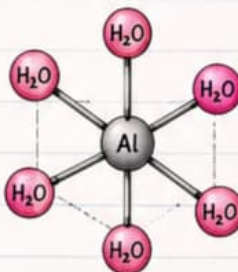
- Atomic size ↑
- Ionization enthalpy ↓
- Metallic character ↑
- Stability of +3 state ↓
- Stability of +1 state ↑

Inert Pair Effect

Reluctance of the valence s-electrons to participate in bonding due to poor shielding by $(n-1)d$ and $(n-2)f$ electrons.

(ii) Aluminium (Al)

- Exhibits amphoteric nature.
- Forms $[Al(H_2O)_6]^{3+}$ in aqueous solution.



$[Al(H_2O)_6]^{3+}$
Octahedral
 $sp^3 d^2$ hybridisation

★ Aluminium is widely used due to its light weight & corrosion resistance.

5. GROUP 14 ELEMENTS

- General configuration: $ns^2 np^2$
- Oxidation states: +4 (most stable), +2 (more stable down the group due to inert pair effect).

Examples:

Element	Important Points	Example/Compound
Carbon	- Tetravalent - Catenation is maximum.	CO_2 (Linear)
Silicon	- Less catenation than carbon. - Forms polymers.	$(SiO_2)_n$ (Silica)
Lead	Shows +2 state predominantly (inert pair effect).	PbO (Basic oxide)

💡 The stability of +2 state increases from C → Pb due to inert pair effect.

Stability of +2 state

$C < Si < Ge < Sn < Pb$

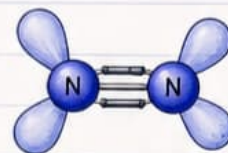
6. GROUP 15 ELEMENTS (Pnictogens)

- General configuration: $ns^2 np^3$
- Oxidation states: -3, +3, +5 (higher for lighter elements; +3 more stable down the group).

Examples:

(i) Nitrogen (N)

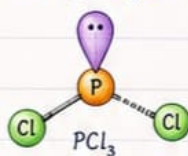
- Diatomic gas (N_2) with $N \equiv N$ triple bond.
- Forms covalent compounds.
- Examples: NH_3 , HNO_3 , N_2O_5



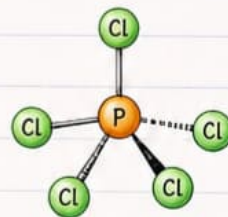
N_2 molecule

(ii) Phosphorus (P)

- Exists as P_4 (white phosphorus).
- Forms PCl_3 (trigonal pyramidal) and PCl_5 (trigonal bipyramidal).



PCl_3
Trigonal pyramidal



PCl_5
Trigonal bipyramidal

Trend down the group:

- Basic character of hydrides ↑
- Reducing character ↑
- Thermal stability of H_3E ↓ ($E = N, P, As, Sb, Bi$)

$H_3N < H_3P < H_3As$
 $< H_3Sb < H_3Bi$



Catenation, multiple oxidation states, and inert pair effect are key features that make p-block elements versatile and important in nature.



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7. GROUP 16 ELEMENTS (Chalcogens)

- General configuration: $ns^2 np^4$
- Oxidation states: -2, +2, +4, +6
- Show catenation.
- Oxides are generally acidic.
- Reactivity: $O > S > Se > Te > Po$

Trends down the group ($O \rightarrow S \rightarrow Se \rightarrow Te \rightarrow Po$)

Atomic size	Increases $\uparrow$
Ionization enthalpy	Decreases $\downarrow$
Electronegativity	Decreases $\downarrow$
Metallic character	Increases $\uparrow$
Stability of -2 state	Decreases $\downarrow$
Stability of +4 state	Increases $\uparrow$
Stability of +6 state	Decreases $\downarrow$

Oxygen - Anomalous Behaviour

- Very small size and high electronegativity.
- Shows only -2, +2 (in OF_2) and not +4, +6.
- Forms $p\pi-p\pi$ multiple bonds (e.g., O_2 , O_3).
- Does not show catenation to a large extent.

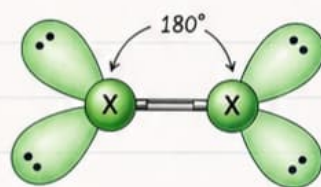
8. GROUP 17 ELEMENTS (Halogens)

- General configuration: $ns^2 np^5$
- Most reactive non-metals.
- Exist as diatomic molecules (X_2).
- Oxidation states: -1 (most stable), +1, +3, +5, +7 (except F)
- Reactivity: $F_2 > Cl_2 > Br_2 > I_2$

Physical Properties Down the Group

Atomic size	Increases $\uparrow$
Colour	Pale yellow $\rightarrow$ Dark brown
Melting point	Increases $\uparrow$
Boiling point	Increases $\uparrow$
Density	Increases $\uparrow$
Reactivity	Decreases $\downarrow$

Structure of Halogen Molecules (X_2)



Each halogen atom forms one σ -bond using p -orbital overlap ($p_z - p_z$).

Oxidising Ability

$F_2 > Cl_2 > Br_2 > I_2$
(Strong oxidising agent)

Interhalogen Compounds

- Formed by combination of two different halogens.
- Examples: ClF , BrF_3 , ICl , IF_7
- Halogens in higher oxidation state behave as Lewis acids.

★ Fluorine shows only -1 oxidation state due to very high electronegativity and small size.

9. GROUP 18 ELEMENTS (Noble Gases)

- General configuration: $ns^2 np^6$ (except He: $1s^2$)
- Chemically inert due to complete valence shell.
- Monoatomic, colourless, odourless gases.
- Very low boiling points.
- Used in bulbs, lasers, balloons, etc.

Physical Properties Down the Group

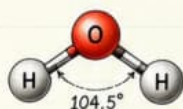
Atomic size	Increases $\uparrow$
Boiling point	Increases $\uparrow$
Density	Increases $\uparrow$
Polarizability	Increases $\uparrow$

Uses

- He - Balloons, cryogenics
- Ne - Sign lights
- Ar - Bulbs, welding
- Kr - Flash lamps
- Xe - High intensity lamps
- Rn - Cancer treatment (Radioisotope)

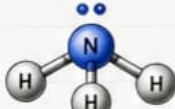
10. IMPORTANT COMPOUNDS OF p-BLOCK ELEMENTS

Water (H_2O)



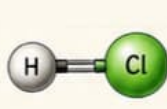
- Bent shape (V-shaped).
- Extensive hydrogen bonding.
- High b.p., high surface tension.
- Universal solvent.

Ammonia (NH_3)



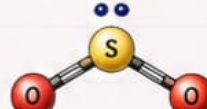
- Trigonal pyramidal.
- Basic in nature.
- Forms hydrogen bonds.
- Highly soluble in water.

Hydrogen Chloride (HCl)



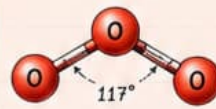
- Covalent molecule.
- Strong acid in water.
- Used in PVC, dyes, medicines.

Sulphur Dioxide (SO_2)



- Bent shape.
- Reducing agent.
- Causes acid rain.
- Used in preserving food.

Ozone (O_3)



- Bent shape.
- Strong oxidising agent.
- Present in stratosphere (ozone layer).

Key Takeaway



p-Block elements show great diversity in properties due to variation in atomic size, electronegativity, and ability to form multiple oxidation states and bonds.

Understand trends today, score high tomorrow!





Chapter-10: The p-Block Elements

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11. COMPARISON WITH s-BLOCK ELEMENTS

Property	s-Block Elements	p-Block Elements
General configuration	ns^{1-2}	$ns^2 np^{1-6}$
Valence electrons	1 or 2	3 to 8
Nature of elements	Highly electropositive (reactive metals)	Metals, non-metals & metalloids (varied nature)
Oxidation states	Only +1 and +2	Many oxidation states (-4 to +7)
Type of compounds	Mainly ionic	Covalent as well as ionic
Catenation	Rare	Very common
Multiple bonding	Rare	Common ($p\pi-p\pi$, $p\pi-d\pi$)
Oxides	Basic oxides	Acidic, basic or amphoteric
Variation in properties	Less variation	Great variation across period

Some Important p-Block Compounds and their Uses

- **Silica (SiO_2)** – Used in glass, ceramics, and as an abrasive.
- **CaCO_3** – Used in cement, antacids, and as a building material.
- **SO_2** – Used in food preservation and in the manufacture of H_2SO_4 .
- **HCl** – Used in cleaning, pickling metals and in PVC production.
- **NH_3** – Used in fertilizers, refrigeration and in cleaning.

12. CHEMICAL BEHAVIOUR – SOME IMPORTANT TRENDS

(i) Acidic Nature of Oxides

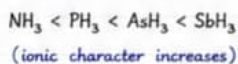
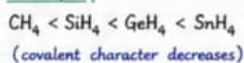
Across a period → increases
Down a group → decreases



(ii) Hydrides

Across a period → More covalent character
Down a group → More ionic character

Example:



(iii) Halides

Across a period → Covalent character increases
Down a group → Ionic character increases

Example (Acid strength):



(iv) Metallic Character

Down a group → Metallic character increases

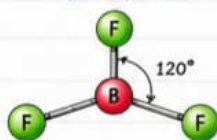
Example (Group 14):

- C (Non-metal)
 - Si (Metalloid)
 - Ge (Metalloid)
 - Sn (Metal)
 - Pb (Metal)
- Metallic character increases ↓

13. STRUCTURES OF SOME IMPORTANT COMPOUNDS

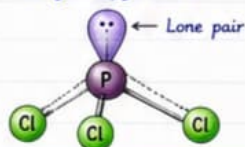
(ii) BF_3 (Boron Trifluoride)

Trigonal planar



(ii) PCl_3 (Phosphorus Trichloride)

Trigonal pyramidal



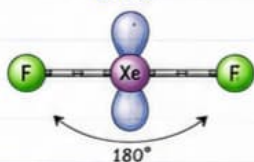
(iii) SF_6 (Sulphur Hexafluoride)

Octahedral



(iv) XeF_2 (Xenon Difluoride)

Linear

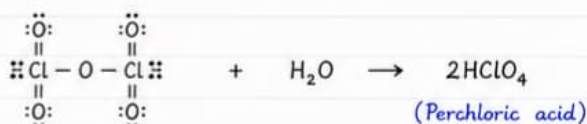


Key Points

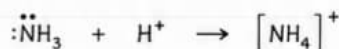
- p-Block elements show wide range of oxidation states.
- Catenation and multiple bonding are prominent.
- Oxides show acidic to amphoteric nature.
- Inert pair effect increases down the group.
- These elements and their compounds are essential in everyday life and industries.

14. EXAMPLES FROM NCERT

(ii) Cl_2O_7 – Strong oxidising agent



(ii) Formation of Ammonium Ion



Quick Revision:

- ✓ Groups 13-18 → last electron in p-orbital.
- ✓ Great variation in physical & chemical properties.
- ✓ Multiple oxidation states (-4 to +7).
- ✓ Catenation, allotropy, multiple bonding → key features.
- ✓ Compounds used in daily life and industries.



★ Master the trends and you will master the p-block!



All the Best
Future Doctor! 😊